

Project 2 : Image compression using SVG

1. What the SVD produces and what the singular values mean, in your own words.

SVD (Singular Value Decomposition) takes a matrix and breaks it into **three simpler matrices**:

$$A = U \Sigma V^T$$

- $U \rightarrow$ tells us about the patterns/directions in the rows.
- Σ (Sigma) $\rightarrow$ tells us **how important each pattern is**.
- $V^T \rightarrow$ tells us about the patterns/directions in the columns.

We use singular values to represent the most important information of the image. These are the singular values of the image. Which therefore results in the compressed image.

2. Your measurement method, including why the storage count is what it is.

I measure compression by comparing the number of scalar values required to represent the original image with the number required by its rank- k SVD representation. The original $m \times n$ image requires mn values, while the truncated SVD requires $mk + k + kn = k(m + n + 1)$ values for U_k , Σ_k , and V_k^T , respectively.

For image size = $mn = 3376 \times 6000 = 20,256,000$

$$mk + k + kn = (3376 \times 50) + 50 + (50 \times 6000) = 468,850$$

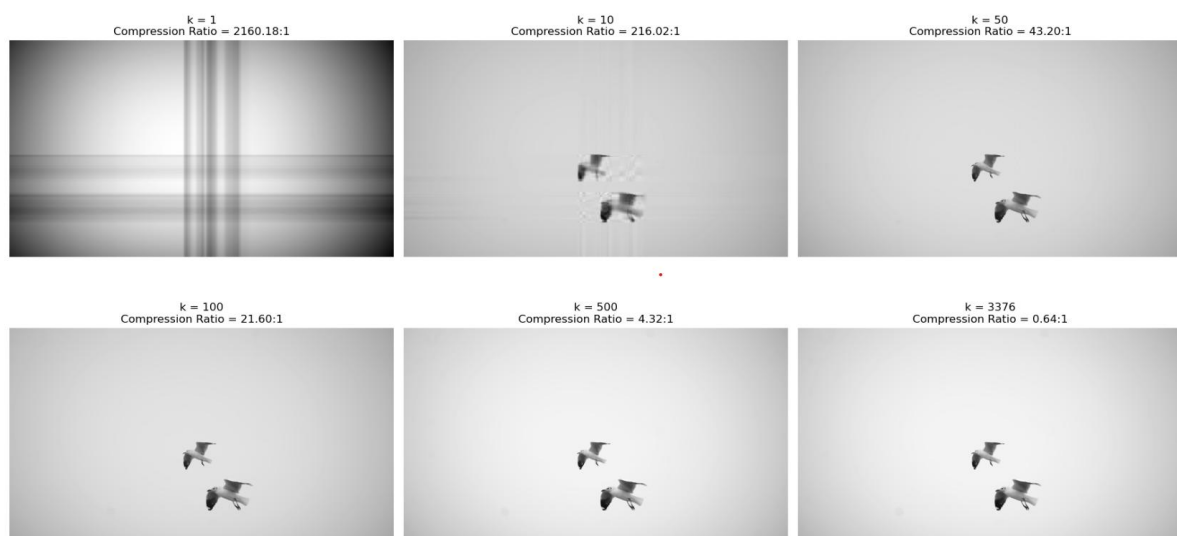
The ratio = 43.21 : 1

Which means With $k = 50$, your SVD representation uses about **43 times fewer values** than the original matrix.

3. The comparison across image types, with the structural explanation.

Using this image :

Picture1 - Truncated SVD Reconstructions



k	Compression ratio	What we see
1	2160.18:1	Almost all detail is lost; only broad background structure remains
10	216.02:1	Birds start appearing, but they are very blurry/distorted
50	43.20:1	Birds become recognizable and most important structure is preserved
100	21.60:1	Much clearer reconstruction
500	4.32:1	Very close to the original
3376	0.64:1	Essentially full-rank reconstruction

4. Your recommended way to choose k , and its limitations.

I used an energy-retention threshold as the primary method because it provides a simple, quantitative, reproducible criterion for selecting the smallest k that preserves a specified amount of the matrix's energy. The elbow method is useful for identifying diminishing returns but can be ambiguous, while a relative-error threshold directly controls reconstruction error but still requires an arbitrary tolerance. The selected k should ultimately be validated using compression ratio and visual quality.

The main limitation is that 95% is an arbitrary threshold. Energy retention does not directly measure visual quality. A small object or important edge can be visually significant while contributing relatively little to the overall energy.

Therefore, after selecting k , I would also check:

1. **Relative error** — how different is the reconstruction?
2. **Compression ratio** — how much storage is saved?
3. **Visual quality** — are important details preserved?

5. When SVD compression is a bad idea, with a reason.

SVD compression is a bad choice when an image contains substantial fine detail, sharp edges, or high-frequency textures that must be preserved accurately. Truncating the SVD discards smaller singular components, which can remove these details and introduce blurring or artifacts. It can also be inefficient when a large number of singular components must be retained, because the SVD representation may require more storage than the original image matrix. Therefore, SVD is most useful when the image has sufficient redundancy or low-rank structure and some loss of detail is acceptable.